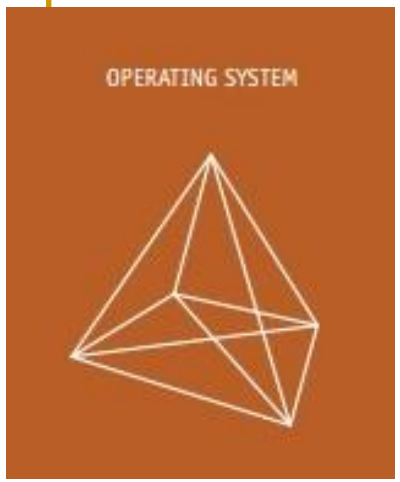
ASTANA IT UNIVERSITY



Operating Systems

Lecture 2 - **The Shell. Command Line Skills. Getting Help**

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Outline

- Shell
- Prompt Description
- Getting Help
- Command Line Skills
- System Information
- Network capabilities
- Login In via SSH

Shell

- The shell is the command interpreter.

- **Features**:

1. Interactive environment: The dialog between the user and the host system.
2. Shell scripts: Linux commands strung together and executed as individual files.
3. I/O redirections: Linux commands can be instructed to take their instructions from files, and not from keyboard. The shell also allows the user to place the output of commands into a file and not on screen of the terminal. Output can also be redirected to other devices, such as to a printer or another terminal on the network.
4. Piping Mechanism: Linux supplies "pipe" facility that allows the output of one command to be used as input in another Linux command.
5. Metacharacter facility: "metacharacters" allows the user to apply selection criteria when accessing files.
6. Background processing: This allows the command to be processed in the background while the user can proceed other task in foreground.

Shell (sh)

- Examples of Shell:
- 1. Bourne again Shell (bash) – 1970s.
- Most common and installed as a default on many Linux distributions
- Widely use for scripting and automation tasks
- Allows to create aliases for commands and customize environment easily
- Includes features from the Cshell and the Korn Shell

- 2. Korn Shell (ksh) – 1993
- Designed as a powerful alternative to the original sh
- Includes features like arithmetic expressions and enhanced string manipulation.
- Provides advanced history substitution mechanisms
- Ksh93, the most recent version, is POSIX-compliant, making it a good for writing portable scripts

- 3. C Shell – early 1980s
- Has a syntax resembling the C programming language
- Better to interactive tasks rather than scripting
- Uses aliases to create command shortcuts
- Advanced history

Prompt Description

The bash prompt:

- It is a string of characters displayed on the terminal screen that indicates that the system is ready for our input.
- It typically includes information such as the current user, the computer's hostname, and the current working directory.
- The prompt is usually displayed on a new line, and the cursor is positioned after the prompt, ready for the user to start typing a command.
- It can be customized to provide useful information to the user. The format can look something like this:

```
<username>@<hostname><current working directory>$
```

```
tosha14@tosha14-VirtualBox:~$
```

Prompt Description

- The **home directory** for a user is marked with a tilde <~> and is the default folder when we log in. The dollar sign, in this case, stands for a user.

```
<username>@<hostname>[~]$
```

- As soon as we log in as **root**, the character changes to a **hash <#>** and looks like this:

```
root@htb[/htb]#
```

Example:
user can shutdown the system in two ways:
$**halt** (log out and shut down)
#**shutdown** -h now (or -r option to shut down and reboot/ +0=now)

Prompt Description

- In addition to providing basic information like the current user and working directory, we can customize to display other information in the prompt, such as the date and time, IP address, the exit status of the last command, and more (table is in the next slide).
- Customizing the prompt can be a useful way to make your terminal experience more personalized and efficient. It can also be a helpful tool for troubleshooting and problem-solving, as it can provide important information about the system's state at any given time.
- In addition to customizing the prompt, we can customize their terminal environment with different color schemes, fonts, and other settings to make their work environment more visually appealing and easier to use (same as in Windows GUI)
- The adjustment of the bash prompt is outside the scope of this module. However, we can look at the bashrcgenerator and powerline, which gives us the possibility to adapt our prompt to our needs.

Prompt Description

Special Character	Description
\d	Date (Mon Feb 6)
\D{%Y-%m-%d}	Date (YYYY-MM-DD)
\H	Full hostname
\j	Number of jobs managed by the shell
\n	Newline
\r	Carriage return
\s	Name of the shell
\t	Current time 24-hour (HH:MM:SS)
\T	Current time 12-hour (HH:MM:SS)
\@	Current time
\u	Current username
\w	Full path of the current working directory

Prompt Description

- You can customize your Bash shell prompt by editing the PS1 (Prompt String 1) variable in your ~/.bashrc file.

  # Customized Bash prompt
  PS1="\u@\h:\w\$ "

  Your prompt will now display something like this:
  your_username@your_hostname:/path/to/current
  /directory$

Unprivileged - User Shell Prompt
$

Privileged - Root Shell Prompt
#

Getting Help

Syntax:

```
tosha14@ubuntu:~$ man <tool>
```

Syntax:

```
tosha14@ubuntu:~$ <tool> --help
```

Syntax:

```
tosha14@ubuntu:~$ <tool> -h
```

Syntax:

```
tosha14@ubuntu:~$ apropos <keyword>
```

Another useful resource to get help if we have issues to understand a long command is: https://explainshell.com/

Command Line Skills

Typical format of the command:

```
command [options] [arguments]
```

ls Lists directory contents
pwd Returns working directory name
cd Changes the directory
touch Creates an empty file
apt High-level package management command-line utility
ifconfig is used to assign or view an address to a network interface and/or configure network interface parameters
sudo Execute command as a different user.
systemctl Command-line based service and systemd control manager
echo Command that displays output in the terminal.
export Command that turns a local variable into an environment variable.
history Command that outputs a list of previously executed commands.
type Command that determines information about command type.

System Information

Command	Description
whoami	Displays current username.
id	Returns users identity
hostname	Sets or prints the name of current host system.
uname	Prints basic information about the operating system name and system hardware.
pwd	Returns working directory name.
ifconfig	The ifconfig utility is used to assign or to view an address to a network interface and/or configure network interface parameters.
ip	Ip is a utility to show or manipulate routing, network devices, interfaces and tunnels.
netstat	Shows network status.
ss	Another utility to investigate sockets.
ps	Shows process status.
who	Displays who is logged in.
env	Prints environment or sets and executes command.
lsblk	Lists block devices.
lsusb	Lists USB devices
lsof	Lists opened files.
lspci	Lists PCI devices.

System Information

- **Hostname**

The hostname command is pretty self-explanatory and will just print the name of the computer that we are logged into

```
tosha14@tosha14-VirtualBox:~$ hostname
tosha14-VirtualBox
```

- **Whoami**

This quick and easy command can be used on both Windows and Linux systems to get our current username.

```
tosha14@tosha14-VirtualBox:~$ whoami
tosha14
```

System Information

- **Id**

The **id** command expands on the whoami command and prints out our effective group membership and IDs. This can be of interest to penetration testers looking to see what access a user may have and sysadmins looking to audit account permissions and group membership. In this output, the group is standard, the adm group means that the user can read log files in /var/log and could potentially gain access to sensitive information, membership in the sudo group is of particular interest as this means our user can run some or all commands as the all-powerful root user. Sudo rights could help us escalate privileges or could be a sign to a sysadmin that they may need to audit permissions and group memberships to remove any access that is not required for a given user to carry out their day-to-day tasks.

```
tosha14@tosha14-VirtualBox:~$ id
uid=1000(tosha14) gid=1000(tosha14) groups=1000(tosha14),4(adm),24(cdrom),27(sud
o),30(dip),46(plugdev),122(lpadmin),135(lxd),136(sambashare)
```

System Information

- **Uname**

Let's dig into the **uname** command a bit more. Check: man uname
Running **uname -a** will print all information about the machine in a specific order: kernel name, hostname, the kernel release, kernel version, machine hardware name, and operating system. The -a flag will omit -p (processor type) and -i (hardware platform) if they are unknown.

```
tosha14@tosha14-VirtualBox:~$ uname -a
Linux tosha14-VirtualBox 6.2.0-32-generic #32~22.04.1-Ubuntu SMP PREEMPT_DYNAMIC
 Fri Aug 18 10:40:13 UTC 2 x86_64 x86_64 x86_64 GNU/Linux
```

System Information

- **Uname** `to Obtain Kernel Release`

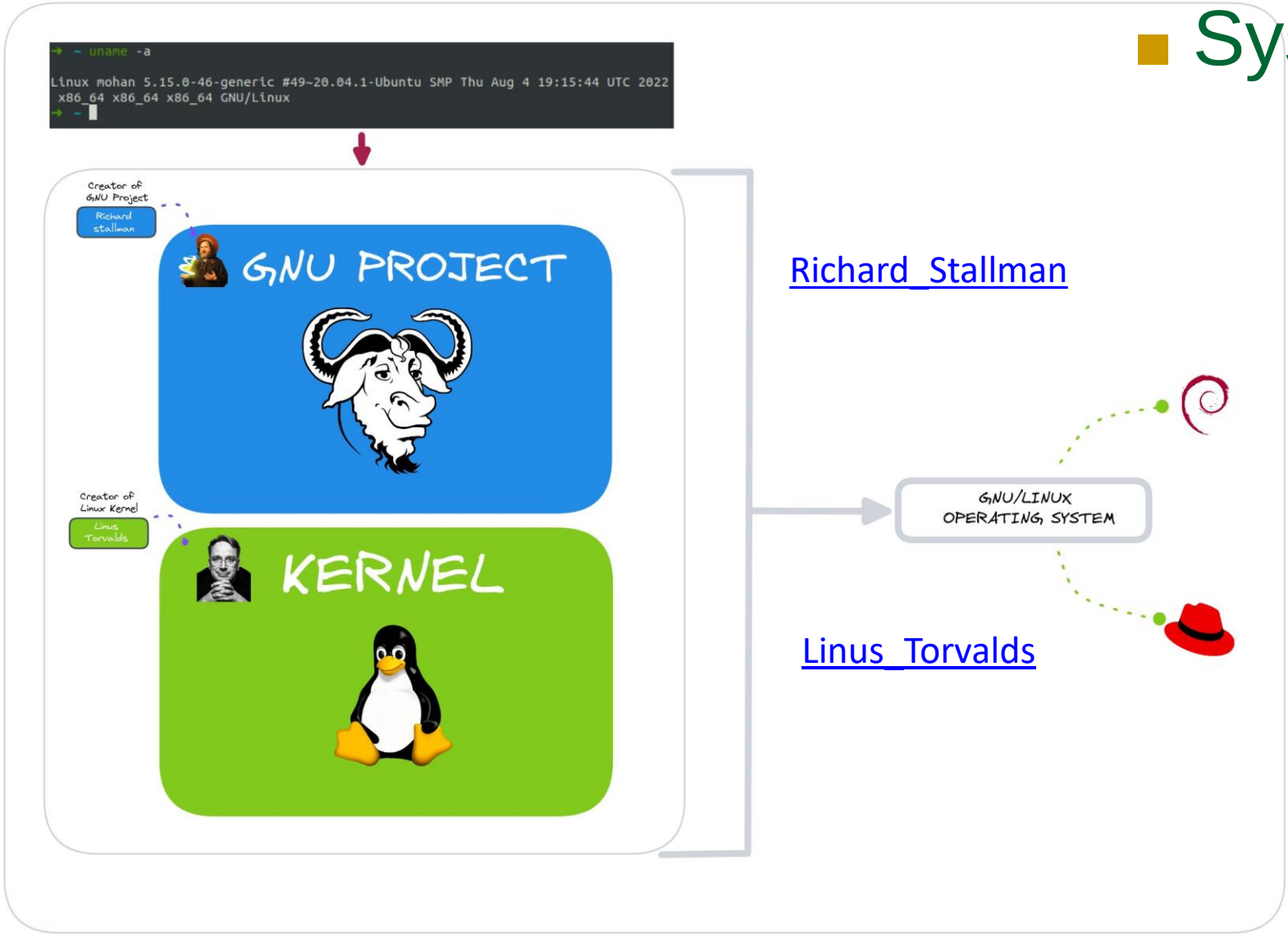
Suppose we want to print out the kernel release to search for potential kernel exploits quickly. We can type uname -r to obtain this information.

```
tosha14@tosha14-VirtualBox:~$ uname -r
6.2.0-32-generic
```

With this info, we could go and search for " 6.2.0-32-generic exploit," and the first result immediately appears useful to us.

- It is highly recommended to study the commands and understand what they are for and what information they can provide. Though a bit tedious, we can learn much from studying the manpages for common commands. We may even find out things that we did not even know were possible with a given command.

System Information



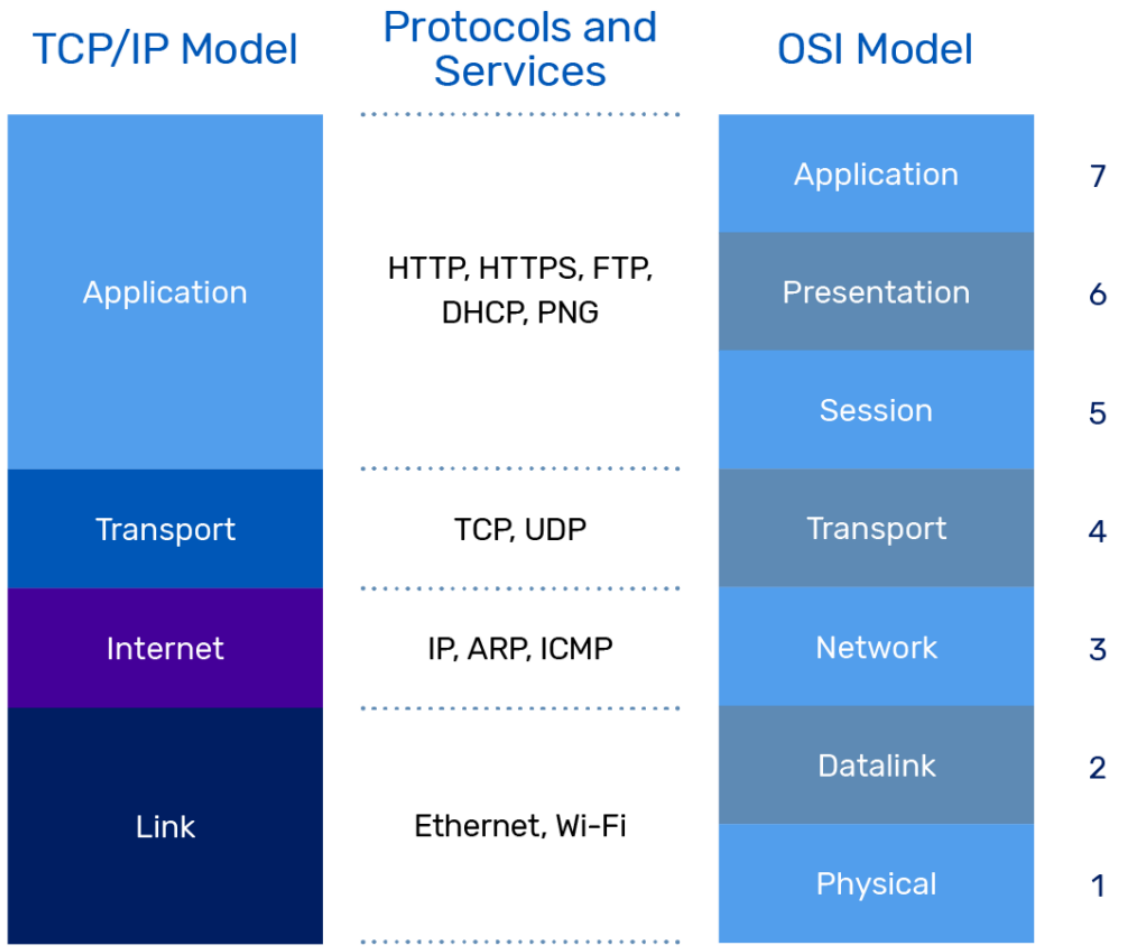
Network capabilities

Ip address (IPv4)

- 4 octets, example: 8.8.8.8, 5.255.0.255
- To identify hosts

Number of port

- Two types of ports: TCP and UDP
- 2 bytes word (from 0 to 65535)
- To identify applications



Logging In via SSH

- **Secure Shell (SSH)** refers to a protocol that allows clients to access and execute commands or actions on remote computers. On Linux-based hosts and servers running or another Unix-like operating system, SSH is one of the permanently installed standard tools and is the preferred choice for many administrators to configure and maintain a computer through remote access. It is an older and very proven protocol that does not require or offer a graphical user interface (GUI). For this reason, it works very efficiently and occupies very few resources. We can connect to our targets with the following command:

```
tosha14@htb[/htb]$ ssh [username]@[IP address]
```

Logging In via SSH

Setting up an **SSH** connection involves two main steps:

- Installing the SSH server
- Connecting to it using SSH client software

Algorithm of installing SSH Server on Kali Linux:

- Update the packages list: sudo apt update
- Install: sudo apt install openssh-server
- Enable: sudo systemctl enable ssh
- Start: sudo systemctl start ssh
- Check the server: sudo systemctl status ssh
- Check the IP of : ifconfig

Algorithm of Connecting to Kali Linux via SSH, so from another computer (Client Machine):

- ssh <username>@<kali-ip>

Logging In via SSH

Troubleshoot:

- Check if IP and username are correct
- Verify Your SSH Configuration
- Check Your Internet Connection

VirtualBox provides several networking modes that you can use to configure how virtual machines (VMs) interact with your physical network and with each other. Each network mode has its own characteristics and use cases. For example, if you need VMs to access the internet but remain somewhat isolated, NAT mode is suitable. If you want VMs to be full members of your network, Bridged mode is often the best choice. For isolated environments, Host-Only or Internal networking may be preferred. Remember that you can configure these settings when creating or editing a VM in VirtualBox, and you can change the network mode at any time to suit your needs.

- Google your problem, use **https://www.8host.com/blog/ustranenie-nepoladok-ssh-problemy-s-podklyucheniem-k-serveru/**

Questions

- Find out the machine hardware name.
- Name ways to get help.
- Which kernel version is installed on the system?
- How to get information about network interfaces?
- What is SSH connection for?